Chapter 9. Rigid Motions of Euclidean Space

Distance-preserving maps, orthogonal matrices and reflections

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Rigid motions of Euclidean space motivate the inner product spaces, groups and forms of later chapters.

9.1 The dot product on $\mathbb{R}^n$

Motivation. The dot product gives $\mathbb{R}^n$ length, distance, angle and perpendicularity.

Definition 9.1.1 (Dot product)

For $X = (a_1, \ldots, a_n)^t$, $Y = (b_1, \ldots, b_n)^t \in \mathbb{R}^n$,

$$\langle X, Y \rangle = X^t \cdot Y = \sum_{i=1}^{n} a_i b_i$$

is the **dot product** on $\mathbb{R}^n$; $\mathbb{R}^n$ with the dot product is **Euclidean space**.

Exercise 9.1.2

(a) $(X, Y) \mapsto \langle X, Y \rangle$ is a bilinear form on $\mathbb{R}^n$ (Definition 6.1.1).
(b) $\langle X, Y \rangle = \langle Y, X \rangle$.
(c) $\langle X, X \rangle \geq 0$, with equality iff $X = 0$.

Solution.

(a) $X^t(aY + bZ) = aX^tY + bX^tZ$, and similarly in X.

(b) $\sum a_i b_i = \sum b_i a_i$.

(c) $\langle X, X \rangle = \sum a_i^2 \geq 0$, and a sum of squares of reals vanishes iff every $a_i = 0$. □

Exercise 9.1.3

If $\langle Y, X \rangle = 0$ for all $Y \in \mathbb{R}^n$, then $X = 0$.

Solution. Take $Y = X$ and use Exercise 9.1.2(c). □

Definition 9.1.4 (Norm)

$\|X\| = \sqrt{\langle X, X \rangle}$ is the (**Euclidean**) **norm**, or length, of X.

Exercise 9.1.5

For $X \in \mathbb{R}^n$, $c \in \mathbb{R}$:
(a) $\|X\| \geq 0$, and $\|X\| = 0$ iff $X = 0$.
(b) $\|cX\| = |c|\|X\|$.
(c) If $X \neq 0$, then $\left\| \frac{1}{\|X\|} X \right\| = 1$. A vector of norm 1 is a **unit vector**.

Solution.

(a) Exercise 9.1.2(c).

(b) $\|cX\|^2 = c^2 \langle X, X \rangle$.

(c) By (b) with $c = 1/\|X\|$. □

Exercise 9.1.6 (Parallelogram law)

$\|X+Y\|^2 + \|X-Y\|^2 = 2(\|X\|^2 + \|Y\|^2)$.

Solution. Expand: $\|X \pm Y\|^2 = \|X\|^2 \pm 2\langle X, Y\rangle + \|Y\|^2$, and add. □

The angle $\theta \in [0, \pi]$ between X and Y is measured in the plane $\langle X\rangle \oplus \langle Y\rangle \cong \mathbb{R}^2$, where

$$\langle X, Y\rangle = \|X\|\|Y\| \cos\theta. \qquad (\angle)$$

Exercise 9.1.7

(a) (Cauchy–Schwarz) $|\langle X, Y\rangle| \le \|X\|\|Y\|$.
(b) (Triangle inequality) $\|X+Y\| \le \|X\| + \|Y\|$.

Solution.

(a) Trivial if $Y = 0$. Otherwise $0 \le \|X - tY\|^2 = \|X\|^2 - 2t\langle X, Y\rangle + t^2\|Y\|^2$ for all t, so the discriminant gives $\langle X, Y\rangle^2 \le \|X\|^2\|Y\|^2$; equality iff X, Y are dependent.

(b) $\|X+Y\|^2 = \|X\|^2 + 2\langle X, Y\rangle + \|Y\|^2 \le (\|X\| + \|Y\|)^2$ by (a). □

So $\|X - Y\|$ is a distance (metric) on $\mathbb{R}^n$:

Exercise 9.1.8

(a) $\|X - X\| = 0$.
(b) $\|X - Y\| = \|Y - X\|$.
(c) $\|X - Y\| \le \|X - Z\| + \|Z - Y\|$.

Solution.

(a) Clear.

(b) Exercise 9.1.5(b) with $c = -1$.

(c) Exercise 9.1.7(b) applied to $(X - Z) + (Z - Y)$. □

Definition 9.1.9 (Perpendicularity)

(a) $X \perp Y$ (X, Y are **perpendicular**, **orthogonal**) if $\langle X, Y\rangle = 0$.
(b) For $S, T \subseteq \mathbb{R}^n$, $S \perp T$ if $\langle X, Y\rangle = 0$ for all $X \in S$, $Y \in T$.

Exercise 9.1.10

The following are equivalent: (1) $X \perp Y$; (2) $\|X+Y\|^2 = \|X\|^2 + \|Y\|^2$ (Pythagoras); (3) $\|X+Y\| = \|X-Y\|$.

Solution. $\|X \pm Y\|^2 = \|X\|^2 + \|Y\|^2 \pm 2\langle X, Y\rangle$. So (2) and (3) both say $\langle X, Y\rangle = 0$. □

Definition 9.1.11 (Orthogonal and orthonormal sets)

(a) Nonzero $X_1, \ldots, X_m \in \mathbb{R}^n$ with $X_i \perp X_j$ for $i \ne j$ form an **orthogonal subset**; if moreover all are unit vectors, an **orthonormal subset**.
(b) For $W \le \mathbb{R}^n$, a basis $\mathfrak{B}$ of W is an **orthogonal basis** if it is an orthogonal subset, and an **orthonormal basis** if it is orthonormal. Normalizing (Exercise 9.1.5(c)) turns an orthogonal basis into an orthonormal one.

Exercise 9.1.12

(a) An orthogonal subset $\{X_1, \ldots, X_m\}$ of $\mathbb{R}^n$ is linearly independent.
(b) For $W \leq \mathbb{R}^n$, a basis $\{X_1, \ldots, X_m\}$ of W is orthonormal iff $\langle X_i, X_j \rangle = \delta_{ij}$.

Solution.

(a) If $\sum c_i X_i = 0$, then $0 = \langle \sum c_i X_i, X_j \rangle = c_j \|X_j\|^2$, so $c_j = 0$.

(b) This is the definition. □

Exercise 9.1.13

The following are orthonormal bases:
(a) the standard basis $\mathcal{E}$ of $\mathbb{R}^n$;
(b) $\{(1,0)^t, (0,-1)^t\}$ of $\mathbb{R}^2$;
(c) $\{(\cos\theta, \sin\theta)^t, (-\sin\theta, \cos\theta)^t\}$ of $\mathbb{R}^2$;
(d) $\{(\cos\theta, \sin\theta, 0)^t, (-\sin\theta, \cos\theta, 0)^t, (0,0,1)^t\}$ of $\mathbb{R}^3$;
(e) $\{\frac{1}{\sqrt{5}}(2,1)^t, \frac{1}{\sqrt{5}}(-1,2)^t\}$ of $\mathbb{R}^2$;
(f) $\{\frac{1}{\sqrt{2}}(-1,1,0)^t, \frac{1}{\sqrt{3}}(1,1,1)^t, \frac{1}{\sqrt{6}}(1,1,-2)^t\}$ of $\mathbb{R}^3$;
(g) $\{(\cos\theta, \sin\theta, 0, 0)^t, (-\sin\theta, \cos\theta, 0, 0)^t, (0, 0, \cos\eta, \sin\eta)^t, (0, 0, -\sin\eta, \cos\eta)^t\}$ of $\mathbb{R}^4$.

Solution. Distinct vectors have dot product 0 and each has norm 1 ($\cos^2 + \sin^2 = 1$); n orthonormal vectors are independent (Exercise 9.1.12(a)), hence a basis. □

Definition 9.1.14 (Orthogonal complement)

For $S \subseteq \mathbb{R}^n$, $S^\perp = \{X \in \mathbb{R}^n : Y \perp X \text{ for all } Y \in S\}$ ("S perp"). For $W \leq \mathbb{R}^n$, $W^\perp$ is the **orthogonal complement** of W.

Observation 9.1.15

Let $S \subseteq \mathbb{R}^n$, $W \leq \mathbb{R}^n$, and $\mathfrak{C}$ a basis of W.
(a) $S^\perp$ is a subspace of $\mathbb{R}^n$.
(b) $S^\perp = \langle S \rangle^\perp$.
(c) $W^\perp = \mathfrak{C}^\perp$.

Proof.

(a) If $S \perp X$ and $S \perp Y$, then $S \perp X + Y$ and $S \perp cX$.

(b) $S \subseteq \langle S \rangle$ gives $\langle S \rangle^\perp \subseteq S^\perp$; conversely $\langle S \rangle$ consists of linear combinations of S (Proposition 3.1.6).

(c) The case $S = \mathfrak{C}$ of (b). ■

Exercise 9.1.16

Let $\mathfrak{B}$ be a basis of $\mathbb{R}^n$ and $W \leq \mathbb{R}^n$.
(a) $0^\perp = \mathbb{R}^n$ and $(\mathbb{R}^n)^\perp = \mathfrak{B}^\perp = 0$.
(b) $W \cap W^\perp = 0$.
(c) $W \leq (W^\perp)^\perp$.

Solution.

(a) Every X is perpendicular to 0; $(\mathbb{R}^n)^\perp = 0$ by Exercise 9.1.3, and $\mathfrak{B}^\perp = (\mathbb{R}^n)^\perp$ by Observation 9.1.15(c).

(b) $X \in W \cap W^\perp$ gives $\langle X, X\rangle = 0$.

(c) Each $X \in W$ is perpendicular to every vector of $W^\perp$. □

Question 9.1.17

If $W \leq \mathbb{R}^n$, is $W = (W^\perp)^\perp$?

There is no direct proof of $(W^\perp)^\perp \leq W$; we compute $\dim W^\perp$.

Example 9.1.18

For $0 \neq Y \in \mathbb{R}^n$, the linear map $\varphi_Y : \mathbb{R}^n \to \mathbb{R}$, $\varphi_Y(X) = \langle Y, X\rangle = Y^t X = L_{Y^t}(X)$, is surjective, since $\varphi_Y(Y) \neq 0$. By the dimension theorem,

$$\dim\langle Y\rangle^\perp = \dim\ker\varphi_Y = n - \dim\operatorname{im}\varphi_Y = n - 1.$$

So $\dim W^\perp = n - \dim W$ holds when $\dim W = 1$.

If $\dim W^\perp = n - \dim W$ for all W, then $\dim(W^\perp)^\perp = \dim W$, so $W = (W^\perp)^\perp$ by Exercise 9.1.16(c) and dimension count, and $\mathbb{R}^n = W \oplus W^\perp$ by Exercise 9.1.16(b) and Observation 7.6.4.

Question 9.1.19

If $W \leq \mathbb{R}^n$, do we always have $\dim W^\perp = n - \dim W$ and $\mathbb{R}^n = W \oplus W^\perp$?

Yes, by the rank theorem (§10.3).

9.2 Rigid motions of $\mathbb{R}^n$

Motivation. Every rigid motion is a translation composed with a *linear* rigid motion, hence a bijection.

Definition 9.2.1 (Rigid motion)

$M : \mathbb{R}^n \to \mathbb{R}^n$ is a **rigid motion** (**isometry**) of $\mathbb{R}^n$ if $\|M(X) - M(Y)\| = \|X - Y\|$ for all $X, Y \in \mathbb{R}^n$.

Example 9.2.2

For $Y \in \mathbb{R}^n$, the **translation** by Y, $T_Y(X) = X + Y$, is a rigid motion; T_Y is a bijection with $(T_Y)^{-1} = T_{-Y}$, and $T_0 = I$.

Example 9.2.3

For $R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$, the **rotation** L_{R_θ} by θ about the origin is a linear rigid motion of $\mathbb{R}^2$ (Exercise 4.3.5); write $L_{R_\theta} = R_\theta$, a bijection with $(R_\theta)^{-1} = R_{-\theta}$, $R_0 = I$.

Example 9.2.4

$S = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$ gives a linear rigid motion $L_S = S$ of $\mathbb{R}^2$ (Exercise 4.3.6), a bijection with $S^{-1} = S$. (It is a reflection; see §9.4.)

Observation 9.2.5

A rigid motion M is injective.

Proof. $M(X) = M(Y)$ gives $\|X - Y\| = 0$, so $X = Y$. ■

Question 9.2.6

Is every rigid motion of $\mathbb{R}^n$ a bijection?

Yes: Corollary 9.2.12.

Exercise 9.2.7

(a) A composite of rigid motions is a rigid motion.
(b) The inverse of a bijective rigid motion is a rigid motion.

Solution.

(a) $\|MN(X) - MN(Y)\| = \|N(X) - N(Y)\| = \|X - Y\|$.

(b) Put $X = M(X')$, $Y = M(Y')$: $\|M^{-1}X - M^{-1}Y\| = \|X' - Y'\| = \|MX' - MY'\| = \|X - Y\|$. □

For a rigid motion M, $L = T_{-M(0)} \circ M$ is a rigid motion with $L(0) = 0$ and $M = T_{M(0)} \circ L$.

Observation 9.2.8

A rigid motion L of $\mathbb{R}^n$ with $L(0) = 0$ satisfies:
(a) $\|L(X)\| = \|X\|$;
(b) $\langle L(X), L(X)\rangle = \langle X, X\rangle$;
(c) $\langle L(X), L(Y)\rangle = \langle X, Y\rangle$;
(d) if $\mathfrak{B}$ is an orthonormal basis of $\mathbb{R}^n$, so is $L(\mathfrak{B})$.

Proof.

(a) Put $Y = 0$ in $\|L(X) - L(Y)\| = \|X - Y\|$.

(b) Square (a).

(c) Square $\|L(X) - L(Y)\| = \|X - Y\|$ and expand: $\langle LX, LX\rangle - 2\langle LX, LY\rangle + \langle LY, LY\rangle = \langle X, X\rangle - 2\langle X, Y\rangle + \langle Y, Y\rangle$; by (b), $\langle LX, LY\rangle = \langle X, Y\rangle$.

(d) If $\mathfrak{B} = \{X_i\}$, then $\langle L(X_i), L(X_j)\rangle = \langle X_i, X_j\rangle = \delta_{ij}$ by (c), so $L(\mathfrak{B})$ is an orthonormal set of n vectors, hence a basis (Exercise 9.1.12). ■

Observation 9.2.9

If the columns $[A]^1, \ldots, [A]^n$ of $A \in \mathfrak{M}_{n,n}(\mathbb{R})$ form an orthonormal basis of $\mathbb{R}^n$, then L_A is a rigid motion.

Proof. For $X = \sum a_i e_i$, $AX = \sum a_i Ae_i = \sum a_i [A]^i$, so $\|AX\|^2 = \sum_{i,j} a_i a_j \delta_{ij} = \sum a_i^2 = \|X\|^2$. Replacing X by $X - Y$ gives $\|AX - AY\| = \|X - Y\|$. ■

Observation 9.2.10

If a rigid motion L with $L(0) = 0$ fixes every vector of a basis $\mathfrak{B} = \{X_i\}$ of $\mathbb{R}^n$, then $L = I$.

Proof. For $X \in \mathbb{R}^n$, $\langle X_i, L(X)\rangle = \langle L(X_i), L(X)\rangle = \langle X_i, X\rangle$ by Observation 9.2.8(c). So $L(X) - X \in \mathfrak{B}^\perp = 0$ (Exercise 9.1.16(a)). ■

Theorem 9.2.11

(a) A rigid motion L of $\mathbb{R}^n$ with $L(0) = 0$ is linear.
(b) Hence every rigid motion M of $\mathbb{R}^n$ is a translation composed with a linear rigid motion.

Proof.

(a) Put $X_j = L(e_j)$, $A = (X_1, \ldots, X_n)$. By Observation 9.2.8(d) $\{X_j\}$ is an orthonormal basis, so A is invertible (Chapter 5). Then $M' = (L_A)^{-1} \circ L$ is a rigid motion (Observation 9.2.9, Exercise 9.2.7) with $M'(0) = 0$, $M'(e_j) = A^{-1}X_j = e_j$, so $M' = I$ by Observation 9.2.10: $L = L_A$.

(b) With $L = T_{-M(0)} \circ M$, L is linear by (a) and $M = T_{M(0)} \circ L$. ■

Second proof of (a). By Observation 9.2.8(c), for all j,

$$\langle L(e_j), L(X+Y)\rangle = \langle e_j, X+Y\rangle = \langle e_j, X\rangle + \langle e_j, Y\rangle = \langle L(e_j), L(X) + L(Y)\rangle.$$

Since $L(\mathcal{E})$ is a basis (Observation 9.2.8(d)), $L(X+Y) - L(X) - L(Y) \in L(\mathcal{E})^\perp = 0$. Likewise $\langle L(e_j), L(cX)\rangle = c\langle e_j, X\rangle = \langle L(e_j), cL(X)\rangle$ gives $L(cX) = cL(X)$. ■

Corollary 9.2.12

Every rigid motion of $\mathbb{R}^n$ is a bijection.

Proof. A linear rigid motion is injective (Observation 9.2.5), hence bijective (dimension theorem); combine with Theorem 9.2.11(b). ■

Exercise 9.2.13

Let $\mathcal{A}(\mathbb{R}^n)$ be the set of rigid motions of $\mathbb{R}^n$. (a) If $L, M \in \mathcal{A}(\mathbb{R}^n)$, then $L \circ M \in \mathcal{A}(\mathbb{R}^n)$.
(b) If $L \in \mathcal{A}(\mathbb{R}^n)$, then $L^{-1} \in \mathcal{A}(\mathbb{R}^n)$.

Solution. Exercise 9.2.7, with Corollary 9.2.12 for (b). □

So $\mathcal{A}(\mathbb{R}^n)$ is a **group** (Chapter 11).

Exercise 9.2.14

Let $L_A, L_B \in \mathcal{A}(\mathbb{R}^n)$ and $Y, Z \in \mathbb{R}^n$.
(a) If $T_Y \circ L_A = T_Z \circ L_B$, then $A = B$ and $Y = Z$. (So every element of $\mathcal{A}(\mathbb{R}^n)$ is uniquely $T_Y \circ L_A$.)
(b) $L_A \circ T_Y = T_{AY} \circ L_A$.

Solution.

(a) Evaluate at 0: $Y = Z$. Then $L_A = L_B$, so $A = B$.

(b) $A(X+Y) = AX + AY$. □

9.3 Orthogonal operators and orthogonal matrices

Motivation. Linear rigid motions are the orthogonal operators, with matrices characterised by $A^tA = I$.

Definition 9.3.1 (Orthogonal operators and matrices)

(a) A *linear* rigid motion of $\mathbb{R}^n$ is a (real) **orthogonal operator**; if $L = L_A$ is one, the real $n \times n$ matrix A is an **orthogonal matrix**.
(b) The **orthogonal group** on $\mathbb{R}^n$ is $\mathbf{O}(\mathbb{R}^n) = \{L \in \mathfrak{L}(\mathbb{R}^n, \mathbb{R}^n) : \|L(X) - L(Y)\| = \|X - Y\| \ \forall X, Y\}$, and

$$\mathbf{O}(n) = \{A \in \mathfrak{M}_{n,n}(\mathbb{R}) : \|AX - AY\| = \|X - Y\| \ \forall X, Y \in \mathbb{R}^n\}.$$

Corollary 9.3.2

For every rigid motion M of $\mathbb{R}^n$ there are unique $A \in \mathbf{O}(n)$, $B \in \mathbb{R}^n$ with $M(X) = AX + B$.

Proof. In Theorem 9.2.11, $M = T_{M(0)} \circ L$ with $L = L_A$ linear; put $B = M(0)$. Uniqueness is Exercise 9.2.14(a). ■

Exercise 9.3.3

(a) If $L, M \in \mathbf{O}(\mathbb{R}^n)$, then $L \circ M \in \mathbf{O}(\mathbb{R}^n)$; if $A, B \in \mathbf{O}(n)$, then $AB \in \mathbf{O}(n)$.
(b) If $L \in \mathbf{O}(\mathbb{R}^n)$, then $L^{-1} \in \mathbf{O}(\mathbb{R}^n)$; if $A \in \mathbf{O}(n)$, then $A^{-1} \in \mathbf{O}(n)$.

Solution. Composites and inverses of linear rigid motions are linear rigid motions (Exercise 9.2.13); $L_AL_B = L_{AB}$ and $(L_A)^{-1} = L_{A^{-1}}$. □

Observation 9.3.4

For linear $L : \mathbb{R}^n \to \mathbb{R}^n$ the following are equivalent:
(1) $\|LX - LY\| = \|X - Y\|$ for all X, Y (i.e. $L \in \mathbf{O}(\mathbb{R}^n)$);
(2) $\|LX\| = \|X\|$ for all X;
(3) $\langle LX, LX\rangle = \langle X, X\rangle$ for all X;
(4) $\langle LX, LY\rangle = \langle X, Y\rangle$ for all X, Y;
(5) if $\mathfrak{B}$ is an orthonormal basis of $\mathbb{R}^n$, so is $L(\mathfrak{B})$.

Proof. (1)⇒(2): $Y = 0$. (2)⇒(1): replace X by $X - Y$, using $L(X - Y) = LX - LY$. (2)⇔(3): square. (3)⇒(4): replace X by $X - Y$ (or $X + Y$) and expand, as in Observation 9.2.8(c). (4)⇒(3): clear. (4)⇒(5): $\langle LX_i, LX_j\rangle = \langle X_i, X_j\rangle = \delta_{ij}$. (5)⇒(2): for an orthonormal basis $\mathfrak{B} = \{X_i\}$ and $X = \sum a_iX_i$, $\|LX\|^2 = \|\sum a_iLX_i\|^2 = \sum a_i^2 = \|X\|^2$. ■

Exercise 9.3.5

The conditions of Observation 9.3.4 are also equivalent to
(2)′ if $\|X\| = 1$, then $\|LX\| = 1$ (L preserves unit vectors);
(5)′ some orthonormal basis $\mathfrak{B}$ has $L\mathfrak{B}$ orthonormal.

Solution. (2)⇒(2)′ and (5)⇒(5)′ are clear. (2)′ ⇒(2): for $X \neq 0$, $\|LX\| = \|X\| \left\|L\left(\frac{X}{\|X\|}\right)\right\| = \|X\|$. (5)′ ⇒(2): the last step of the proof of Observation 9.3.4 uses only one orthonormal basis. □

Observation 9.3.6

For $A \in \mathfrak{M}_{n,n}(\mathbb{R})$ the following are equivalent:
(1) $\|AX - AY\| = \|X - Y\|$ for all X, Y (i.e. $A \in \mathbf{O}(n)$);
(2) $\|AX\| = \|X\|$;
(3) $\langle AX, AX\rangle = \langle X, X\rangle$;
(4) $\langle AX, AY\rangle = \langle X, Y\rangle$;
(5) the columns $\{[A]^1, \dots, [A]^n\}$ form an orthonormal basis of $\mathbb{R}^n$;
(6) $A^tA = I = AA^t$, i.e. $A^{-1} = A^t$;
(7) $A^t \in \mathbf{O}(n)$;
(8) the rows $\{[A]_1, \dots, [A]_n\}$ form an orthonormal basis of $\mathfrak{M}_{1,n}(\mathbb{R})$.

Proof. (1)⇔(2)⇔(3)⇔(4)⇔(5): Observation 9.3.4 for $L = L_A$ with $\mathfrak{B} = \mathcal{E}$, using $[A]^i = Ae_i$.

(5)⇔(6): $(A^tA)_{ij} = \langle [A]^i, [A]^j\rangle$; for square A, $A^tA = I$ implies $AA^t = I$.

(6)⇒(7) [Supplement]: $(A^t)^tA^t = AA^t = I$, so A^t satisfies (6). (7)⇒(6): apply (6) to A^t.

(7)⇔(8) [Supplement]: the rows of A are the transposed columns of A^t; apply (5) to A^t. ■

Second proof of (4)⇔(6). (4) says $X^tY = (AX)^tAY = X^t(A^tA)Y = X^tIY$ for all X, Y, which by Exercise 9.3.7 means $A^tA = I$. The converse reads the computation backwards. ■

Exercise 9.3.7

If $B, C \in \mathfrak{M}_{m,n}(F)$ and $X^tBY = X^tCY$ for all $X \in F^m$, $Y \in F^n$, then $B = C$.

Solution. $e_i^tBe_j = B_{ij}$; take $X = e_i$, $Y = e_j$. □

Hence

$$\mathbf{O}(n) = \{A \in \mathfrak{M}_{n,n}(\mathbb{R}) : A^tA = I\},$$

and with $(L_A)^t = L_{A^t}$, $\mathbf{O}(\mathbb{R}^n) = \{L : L^t \circ L = I\}$.

Exercise 9.3.8

If $A \in \mathbf{O}(n)$, then $\det A = \pm 1$.

Solution. $1 = \det(A^tA) = \det(A^t)\det A = (\det A)^2$. □

Definition 9.3.9 (Special orthogonal group)

$\mathbf{SO}(n) = \{A \in \mathbf{O}(n) : \det A = 1\}$ and $\mathbf{SO}(\mathbb{R}^n) = \{L \in \mathbf{O}(\mathbb{R}^n) : \det L = 1\}$.

L_A is **orientation preserving** if $A \in \mathbf{SO}(n)$, **orientation reversing** if $A \in \mathbf{O}(n) - \mathbf{SO}(n)$.

Exercise 9.3.10

(a) $I \in \mathbf{SO}(n)$.
(b) If $A, B \in \mathbf{SO}(n)$, then $AB \in \mathbf{SO}(n)$.
(c) If $A \in \mathbf{SO}(n)$, then $A^{-1} \in \mathbf{SO}(n)$.

Solution. Exercise 9.3.3 together with $\det(AB) = \det A \det B$ and $\det A^{-1} = (\det A)^{-1}$. □

Exercise 9.3.11

Fix $S \in \mathbf{O}(n) - \mathbf{SO}(n)$.
(a) Indeed $\mathbf{O}(n) - \mathbf{SO}(n) \neq \emptyset$.
(b) $\lambda_S : \mathbf{SO}(n) \to S \cdot \mathbf{SO}(n) = \{SA : A \in \mathbf{SO}(n)\}$, $\lambda_S(A) = SA$, is a bijection.
(c) $\mathbf{O}(n) - \mathbf{SO}(n) = S \cdot \mathbf{SO}(n)$.
(d) $\mathbf{O}(n) = \mathbf{SO}(n) \amalg (S \cdot \mathbf{SO}(n))$ (disjoint union).

Solution.

(a) $\mathrm{diag}(-1, 1, \ldots, 1)$.

(b) $SA = SB$ implies $A = B$; surjective by definition.

(c) If $R \in \mathbf{O}(n) - \mathbf{SO}(n)$, then $S^{-1}R \in \mathbf{SO}(n)$ ($\det = (-1)(-1)$), so $R = S(S^{-1}R)$; conversely $\det(SA) = -1$.

(d) Combine (c) with $\mathbf{SO}(n) \cap (\mathbf{O}(n) - \mathbf{SO}(n)) = \emptyset$. □

9.4 Reflections

Motivation. In an orthonormal frame starting with $Y/\|Y\|$, the reflection in $\langle Y \rangle^\perp$ is $\mathrm{diag}(-1, 1, \ldots, 1)$.

Example 9.4.1

For $X, Y \in \mathbb{R}^3$, $Y \neq 0$, write $X = aY + (X - aY)$ with $(X - aY) \perp Y$; then $a\langle Y, Y\rangle = \langle X, Y\rangle$, so $a = \frac{\langle X,Y\rangle}{\langle Y,Y\rangle}$. Reflecting X in the plane $\langle Y \rangle^\perp$ gives $X - \frac{2\langle X,Y\rangle}{\langle Y,Y\rangle}Y$.

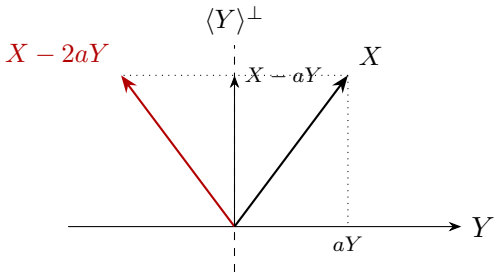


Figure. *Reflection of X in the hyperplane $\langle Y \rangle^\perp$.*

Definition 9.4.2 (Reflection)

For $0 \neq Y \in \mathbb{R}^n$, $S_Y : \mathbb{R}^n \to \mathbb{R}^n$,

$$S_Y(X) = X - \frac{2\langle X, Y\rangle}{\langle Y, Y\rangle}Y,$$

is the **reflection** (**symmetry**) with respect to Y.

Remark 9.4.3

The "reflection in the x-axis" of $\mathbb{R}^2$ is $S_{(0,1)^t}$ in this terminology.

Exercise 9.4.4

For $0 \neq Y \in \mathbb{R}^n$:
(a) $S_Y = S_{cY}$ for $0 \neq c \in \mathbb{R}$;
(b) $S_Y(Y) = -Y$;
(c) if $X \perp Y$, i.e. $X \in \langle Y \rangle^\perp$, then $S_Y(X) = X$.

Solution.
(a) $\frac{\langle X,cY\rangle}{\langle cY,cY\rangle} cY = \frac{c^2\langle X,Y\rangle}{c^2\langle Y,Y\rangle} Y$.
(b) $Y - 2Y = -Y$.
(c) $\langle X, Y\rangle = 0$. □

Exercise 9.4.5

$S_Y \in \mathbf{O}(\mathbb{R}^n)$.

Solution. S_Y is linear because $X \mapsto \langle X, Y\rangle$ is. And $\|S_Y X\|^2 = \|X\|^2 - \frac{4\langle X,Y\rangle^2}{\langle Y,Y\rangle} + \frac{4\langle X,Y\rangle^2}{\langle Y,Y\rangle^2}\langle Y, Y\rangle = \|X\|^2$. Apply Observation 9.3.4. □

By Example 9.1.18 and Observation 7.6.4, $\mathbb{R}^n = \langle Y\rangle \oplus \langle Y\rangle^\perp$. For a basis $\mathfrak{C} = \{X_2, \ldots, X_n\}$ of $\langle Y\rangle^\perp$, $\mathfrak{B} = \{Y, X_2, \ldots, X_n\}$ is a basis of $\mathbb{R}^n$, and Exercise 9.4.4(b),(c) give

$$[S_Y]^{\mathfrak{B}}_{\mathfrak{B}} = \operatorname{diag}(-1, 1, \ldots, 1).$$

Exercise 9.4.6

For $0 \neq Y \in \mathbb{R}^n$, $(S_Y)^2 = I$ and $(S_Y)^{-1} = S_Y$.

Solution. $[S_Y^2]^{\mathfrak{B}}_{\mathfrak{B}} = \operatorname{diag}(-1, 1, \ldots, 1)^2 = I$. □

Exercise 9.4.7

For $0 \neq Y \in \mathbb{R}^n$, $\det(S_Y) = -1$.

Solution. $\det S_Y = \det[S_Y]^{\mathfrak{B}}_{\mathfrak{B}} = \det \operatorname{diag}(-1, 1, \ldots, 1) = -1$. □

As a matrix, $S_Y = L_A$ for a unique $A \in \mathbf{O}(n)$, and

$$A = [S_Y]^{\mathcal{E}}_{\mathcal{E}} = [I]^{\mathfrak{B}}_{\mathcal{E}} \cdot [S_Y]^{\mathfrak{B}}_{\mathfrak{B}} \cdot [I]^{\mathcal{E}}_{\mathfrak{B}} = [I]^{\mathfrak{B}}_{\mathcal{E}} \cdot \operatorname{diag}(-1, 1, \ldots, 1) \cdot [I]^{\mathcal{E}}_{\mathfrak{B}}.$$

$[I]^{\mathcal{E}}_{\mathfrak{B}} \in \mathbf{O}(n)$ iff $\mathfrak{B}$ is orthonormal (Observation 9.3.6(5)); with $\|Y\| = 1$ (Exercise 9.4.4(a)) this means $\mathfrak{C}$ is orthonormal.

Question 9.4.8

Does every $W \leq \mathbb{R}^n$ have an orthonormal basis?

Yes (Gram–Schmidt, next chapter). Then all three factors lie in $\mathbf{O}(n)$ ($B = U^{-1}AU$ with $U \in \mathbf{O}(n)$: *orthogonally similar*): in the frame $\mathfrak{B}$, S_Y flips the Y-axis and fixes the others.

9.5 $\mathbf{O}(2)$ and $\mathbf{SO}(2)$

Motivation. Every orthogonal operator of the plane is a rotation or a reflection, according to its determinant.

Example 9.5.1

$\mathbf{O}(1) = \{1, -1\}$ and $\mathbf{SO}(1) = \{1\}$.

Theorem 9.5.2

The elements of $\mathbf{SO}(2)$ are exactly the 2-dimensional rotations:

$$\mathbf{SO}(2) = \left\{ \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} : 0 \le \theta < 2\pi \right\} = \left\{ \begin{pmatrix} x & -y \\ y & x \end{pmatrix} : x, y \in \mathbb{R},\ x^2 + y^2 = 1 \right\}.$$

High-school proof. Let $A = \left(\begin{smallmatrix} x & z \\ y & w \end{smallmatrix}\right) \in \mathbf{SO}(2)$: $x^2 + y^2 = 1$, $z^2 + w^2 = 1$, $xz + yw = 0$, $xw - yz = 1$. Write $(x, y) = (\cos\theta, \sin\theta)$ and $(z, w) = (\sin\eta, \cos\eta)$. Then $\sin(\theta + \eta) = xz + yw = 0$ and $\cos(\theta + \eta) = xw - yz = 1$, so $\theta + \eta \in 2\pi\mathbb{Z}$. Hence $z = \sin\eta = -\sin\theta$ and $w = \cos\eta = \cos\theta$. ■

Freshman's proof. If $y = 0$, then $x = \pm 1$; $xz = 0$ gives $z = 0$, and $xw = 1$ gives $w = x$: $A = \pm I$, of the stated form. If $y \neq 0$, $w = -\frac{xz}{y}$, and $1 = z^2 + w^2 = \frac{z^2(y^2+x^2)}{y^2} = \frac{z^2}{y^2}$, so $z = \pm y$. If $z = y$, then $w = -x$ and $xw - yz = -(x^2 + y^2) = -1$, impossible; so $z = -y$ and $w = x$. ■

Sophomore's proof. The columns are an orthonormal basis (Observation 9.3.6(5)): $(\cos\theta, \sin\theta)^t$ and then $(-\sin\theta, \cos\theta)^t$ (determinant 1) or $(\sin\theta, -\cos\theta)^t$ (determinant -1). ■

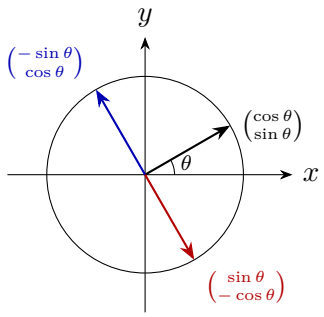


Figure. *The two choices of second basis vector: determinant $+1$ or -1.*

Corollary 9.5.3

$$\mathbf{O}(2) = \mathbf{SO}(2) \amalg \left\{ \begin{pmatrix} x & y \\ y & -x \end{pmatrix} : x, y \in \mathbb{R},\ x^2 + y^2 = 1 \right\}.$$

Proof. By Exercise 9.3.11, $\mathbf{O}(2) - \mathbf{SO}(2) = S \cdot \mathbf{SO}(2)$ for any $S \in \mathbf{O}(2) - \mathbf{SO}(2)$. With $S = \left(\begin{smallmatrix} -1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$, the elements are $S\left(\begin{smallmatrix} x & -y \\ y & x \end{smallmatrix}\right) = \left(\begin{smallmatrix} -x & y \\ y & x \end{smallmatrix}\right)$; replace x by $-x$. ■

Write $R_\theta = \left(\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right)$ and $S_\theta = \left(\begin{smallmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{smallmatrix}\right)$.

Question 9.5.4

Is S_θ a reflection of $\mathbb{R}^2$?

Question 9.5.5

For reflections S_Y, S_Z of $\mathbb{R}^2$, $\det(S_Y S_Z) = (-1)(-1) = 1$, so $S_Y \circ S_Z = R_\theta$ for some θ (Theorem 9.5.2). How do we find θ?

Observation 9.5.6

For $\theta \in \mathbb{R}$ and $Y_\theta = (-\sin\frac{\theta}{2}, \cos\frac{\theta}{2})^t$, $S_\theta = S_{Y_\theta}$. In particular S_θ is a reflection.

Proof. Compare on e_1, e_2, with $s = \sin\frac{\theta}{2}$, $c = \cos\frac{\theta}{2}$: $S_{Y_\theta}(e_1) = e_1 - 2\langle e_1, Y_\theta\rangle Y_\theta = e_1 + 2s(-s,c)^t = (1-2s^2, 2sc)^t = (\cos\theta, \sin\theta)^t = S_\theta e_1$, and $S_{Y_\theta}(e_2) = e_2 - 2c(-s,c)^t = (2sc, 1-2c^2)^t = (\sin\theta, -\cos\theta)^t = S_\theta e_2$ (half-angle formulas). ■

Theorem 9.5.7

An orthogonal operator L of $\mathbb{R}^2$ is a rotation or a reflection: a rotation if $\det L = 1$, a reflection if $\det L = -1$.

Proof. $\mathbf{O}(2) = \mathbf{SO}(2) \amalg (\mathbf{O}(2) - \mathbf{SO}(2)) = \{R_\theta\} \amalg \{S_\theta\}$ (Corollary 9.5.3), and each S_θ is a reflection (Observation 9.5.6). ■

Example 9.5.8

(a) For a rotation R and a reflection S of $\mathbb{R}^2$, $R \circ S$ and $S \circ R$ are reflections (determinant -1).
(b) For reflections S, S', $S \circ S'$ is a rotation (determinant 1).

$R_\theta \circ R_\eta = R_{\theta+\eta}$, $(R_\theta)^{-1} = R_{-\theta}$, and $(S_\theta)^2 = I$ (Exercise 9.4.6).

Observation 9.5.9

For a reflection S of $\mathbb{R}^2$ and $\theta \in \mathbb{R}$: $S^{-1} \circ R_\theta \circ S = (R_\theta)^{-1} = R_{-\theta}$, i.e. $R_\theta \circ S = S \circ R_{-\theta}$, or $(S \circ R_\theta)^2 = I$.

Proof. $S \circ R_\theta$ is a reflection (Example 9.5.8), so $(S \circ R_\theta)^2 = I$ by Exercise 9.4.6. Thus $SR_\theta SR_\theta = I$, i.e. $SR_\theta S = (R_\theta)^{-1} = R_{-\theta}$, and $S = S^{-1}$. ■

Example 9.5.10

Fix a reflection S of $\mathbb{R}^2$. By Exercise 9.3.11, $\mathbf{O}(2) - \mathbf{SO}(2) = S \cdot \mathbf{SO}(2)$, so every reflection is uniquely $S \circ R_\theta$, and Observation 9.5.9 moves S past rotations in any product; e.g. $(S \circ R_\theta) \circ (S \circ R_\eta) = S \circ (S \circ R_{-\theta}) \circ R_\eta = R_{\eta-\theta}$.

Example 9.5.11

Answer to Question 9.5.5: find θ, η with $S_Y = S_\theta$, $S_Z = S_\eta$, i.e. $\frac{1}{\|Y\|}Y = \pm(-\sin\frac{\theta}{2}, \cos\frac{\theta}{2})^t$ and $\frac{1}{\|Z\|}Z = \pm(-\sin\frac{\eta}{2}, \cos\frac{\eta}{2})^t$ (Observation 9.5.6, Exercise 9.4.4(a)). With $S = \left(\begin{smallmatrix} -1 & 0 \\ 0 & 1 \end{smallmatrix}\right) = S_\pi$ we have $S \circ R_{\pi-\theta} = S_\theta$, so by Observation 9.5.9

$$S_\theta \circ S_\eta = (S \circ R_{\pi-\theta}) \circ (S \circ R_{\pi-\eta}) = S^2 \circ R_{\theta-\pi} \circ R_{\pi-\eta} = R_{\theta-\eta}.$$

Exercise 9.5.12

For $Y = (-1, 1)^t$, $Z = (-1, \sqrt{3})^t$, find θ with $S_Y \circ S_Z = R_\theta$.

Solution. $\frac{1}{\|Y\|}Y = (-\sin\frac{\pi}{4}, \cos\frac{\pi}{4})^t$ and $\frac{1}{\|Z\|}Z = (-\frac{1}{2}, \frac{\sqrt{3}}{2})^t = (-\sin\frac{\pi}{6}, \cos\frac{\pi}{6})^t$, so $S_Y = S_{\pi/2}$, $S_Z = S_{\pi/3}$ and $S_Y \circ S_Z = R_{\pi/2-\pi/3} = R_{\pi/6}$. □

Exercise 9.5.13

$R_\theta \circ S_\eta = S_{\theta+\eta}$ and $S_\eta \circ R_\theta = S_{\eta-\theta}$.

Solution. Addition formulas, column by column: $R_\theta S_\eta e_1 = R_\theta(\cos\eta, \sin\eta)^t = (\cos(\theta+\eta), \sin(\theta+\eta))^t$ and $S_\eta R_\theta e_1 = S_\eta(\cos\theta, \sin\theta)^t = (\cos(\eta-\theta), \sin(\eta-\theta))^t$; likewise for e_2. □

Exercise 9.5.14

For a reflection S of $\mathbb{R}^2$, compute $(S \circ R_{\pi/6} \circ S)^{2000}$.

Solution. $S \circ R_{\pi/6} \circ S = R_{-\pi/6}$ (Observation 9.5.9, $S = S^{-1}$). So the power is $R_{-2000\pi/6} = R_{-1000\pi/3}$, and $-\frac{1000\pi}{3} \equiv \frac{2\pi}{3} \pmod{2\pi}$ ($1000 = 3 \cdot 334 - 2$). The answer is $R_{2\pi/3}$. □

Exercise 9.5.15

Every element of $\mathbf{SO}(2)$ is a product of two reflections, but not uniquely.

Solution. By Example 9.5.11, $S_{\theta+\alpha} \circ S_\alpha = R_\theta$ for every $\alpha \in \mathbb{R}$; different $\alpha \pmod{2\pi}$ give different factorizations. □

9.6 $\mathbf{SO}(3)$ and $\mathbf{SO}(n)$

Motivation. Every element of $\mathbf{SO}(3)$ fixes an axis and rotates the perpendicular plane.

Definition 9.6.1 (Rotation of $\mathbb{R}^3$)

Let $Z \in \mathbb{R}^3$ be a unit vector and $\mathfrak{B} = \{Z, X, Y\}$ an orthonormal basis of $\mathbb{R}^3$ (so $\{X, Y\}$ is an orthonormal basis of $\langle Z\rangle^\perp$). Then $(Z, X, Y) \in \mathbf{O}(3)$ has determinant ± 1; order X, Y so that $\det(Z, X, Y) = \det(X, Y, Z) = 1$. The **rotation** $R_{Z,\theta} : \mathbb{R}^3 \to \mathbb{R}^3$ is the linear map with

$$[R_{Z,\theta}]^{\mathfrak{B}}_{\mathfrak{B}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix} = \operatorname{diag}(1, R_\theta);$$

Z is the **axis**, $\langle X\rangle \oplus \langle Y\rangle$ the **plane of rotation**, θ the **angle**.

It does not depend on $\{X, Y\}$ (Exercise 9.6.6). As in §9.4,

$$[R_{Z,\theta}]^{\mathcal{E}}_{\mathcal{E}} = [I]^{\mathfrak{B}}_{\mathcal{E}} \cdot \operatorname{diag}(1, R_\theta) \cdot [I]^{\mathcal{E}}_{\mathfrak{B}},$$

where $\operatorname{diag}(1, R_\theta) \in \mathbf{SO}(3)$ and $[I]^{\mathcal{E}}_{\mathfrak{B}} = (Z, X, Y) \in \mathbf{SO}(3)$.

Observation 9.6.2

Rotations $R_{Z,\theta}$ of $\mathbb{R}^3$ belong to $\mathbf{SO}(\mathbb{R}^3)$.

Proof. By the displayed product all three factors lie in $\mathbf{SO}(3)$ (Exercise 9.3.10). ■

Question 9.6.3

Is the composite of two rotations of $\mathbb{R}^3$ (with different axes) a rotation?

Theorem 9.6.4

Every element of $\mathbf{SO}(\mathbb{R}^3)$ is a rotation of $\mathbb{R}^3$.

Proof. (i) *An axis.* Let $L_A \in \mathbf{SO}(\mathbb{R}^3)$. We need a unit vector Z with $AZ = Z$, i.e. $\det(A - I) = 0$. Since $\det A = \det A^t = 1$ and $A^t A = I$,

$$\begin{aligned}\det(A - I) &= \det(A^t)\det(A - I) = \det(A^t(A - I)) = \det(I - A^t) \\ &= \det((I - A)^t) = \det(I - A) = (-1)^3 \det(A - I),\end{aligned}$$

so $\det(A - I) = 0$.

(ii) *The plane.* For an orthonormal $\mathfrak{B} = \{Z, X, Y\}$ with $\det(Z, X, Y) = 1$, $[L_A]^{\mathfrak{B}}_{\mathfrak{B}} = [I]^{\mathcal{E}}_{\mathfrak{B}} A [I]^{\mathfrak{B}}_{\mathcal{E}} = \left(\begin{smallmatrix} 1 & a & b \\ 0 & & \\ & B & \end{smallmatrix}\right) \in \mathbf{SO}(3)$. Orthonormal columns force $a = b = 0$, $B \in \mathbf{O}(2)$, and $\det B = 1$, so $B = R_\theta$ (Theorem 9.5.2). ■

Second proof of (i). If $A \in \mathbf{O}(3)$ has a real eigenvalue λ with eigenvector X, then $\langle X, X\rangle = \langle AX, AX\rangle = \lambda^2\langle X, X\rangle$, so $\lambda = \pm 1$. Triangularizing A over $\mathbb{C}$ (§7.3), its eigenvalues satisfy $\lambda_1\lambda_2\lambda_3 = \det A = 1$ (Corollary 7.3.3). A real cubic has a real root, say $\lambda_1 = \pm 1$. If $\lambda_2, \lambda_3 \notin \mathbb{R}$, then $\lambda_3 = \bar{\lambda}_2$ and $\lambda_2\lambda_3 = |\lambda_2|^2 > 0$, so $\lambda_1 = 1$. If all are real, they are ± 1 with product 1, so one of them is 1. ■

Corollary 9.6.5

The composite of two rotations of $\mathbb{R}^3$ is a rotation.

Proof. By Observation 9.6.2 and Exercise 9.3.10 the composite lies in $\mathbf{SO}(\mathbb{R}^3)$; apply Theorem 9.6.4. ■

The axis of $L_{A_1} \circ L_{A_2}$ is the unit eigenvector of $A_1 A_2$ for eigenvalue 1.

Exercise 9.6.6

Let $\mathfrak{B}' = \{Z, X', Y'\}$ be another orthonormal basis with $\det(Z, X', Y') = 1$, and $R'_{Z,\theta}$ the rotation defined from it. Show $R_{Z,\theta} = R'_{Z,\theta}$:
(a) since $\langle X\rangle \oplus \langle Y\rangle = \langle X'\rangle \oplus \langle Y'\rangle = \langle Z\rangle^\perp$, $[I]^{\mathfrak{B}}_{\mathfrak{B}'} = \operatorname{diag}(1, U)$ for some $U \in \mathfrak{M}_{2,2}(\mathbb{R})$;
(b) $[I]^{\mathfrak{B}}_{\mathfrak{B}'} = [I]^{\mathfrak{B}}_{\mathcal{E}}[I]^{\mathcal{E}}_{\mathfrak{B}'} \in \mathbf{SO}(3)$, hence $U \in \mathbf{SO}(2)$;
(c) $[R'_{Z,\theta}]^{\mathfrak{B}}_{\mathfrak{B}} = [I]^{\mathfrak{B}}_{\mathfrak{B}'} \operatorname{diag}(1, R_\theta)[I]^{\mathfrak{B}'}_{\mathfrak{B}} = \operatorname{diag}(1, R_\theta) = [R_{Z,\theta}]^{\mathfrak{B}}_{\mathfrak{B}}$.

Solution.

(a) Z is fixed and $X', Y' \in \langle X, Y\rangle$, so the change of coordinates has this block form.

(b) Both factors are in $\mathbf{SO}(3)$ (columns orthonormal, determinant 1), so $\operatorname{diag}(1, U) \in \mathbf{SO}(3)$, whence $U \in \mathbf{O}(2)$ with $\det U = 1$.

(c) $\operatorname{diag}(1, U)\operatorname{diag}(1, R_\theta)\operatorname{diag}(1, U^{-1}) = \operatorname{diag}(1, UR_\theta U^{-1})$, and $UR_\theta U^{-1} = R_\theta$ because $\mathbf{SO}(2)$ is commutative ($R_\alpha R_\theta = R_{\alpha+\theta}$). □

Example 9.6.7

Unlike Theorem 9.5.7: for $R_\theta \neq I$, $\operatorname{diag}(-1, R_\theta) \in \mathbf{O}(3)$ has determinant -1 (not a rotation) and no 2-dimensional 1-eigenspace (not a reflection).

Exercise 9.6.8

For $L \in \mathbf{O}(\mathbb{R}^3) - \mathbf{SO}(\mathbb{R}^3)$:
(a) L has eigenvalue -1;
(b) $[L]^{\mathfrak{B}}_{\mathfrak{B}} = \operatorname{diag}(-1, R_\theta)$ for some orthonormal basis $\mathfrak{B}$ and $\theta \in \mathbb{R}$.

Solution.

(a) With $L = L_A$, $\det A = -1$: $\det(A + I) = -\det(A^t)\det(A + I) = -\det(I + A^t) = -\det(A + I)$, so $\det(A + I) = 0$.

(b) Take a unit eigenvector Z for -1 and an orthonormal basis $\{Z, X, Y\}$. As in Theorem 9.6.4(ii), $[L]^{\mathfrak{B}}_{\mathfrak{B}} = \operatorname{diag}(-1, B)$ with $B \in \mathbf{O}(2)$ and $\det B = 1$, so $B = R_\theta$. □

So in a suitable orthonormal frame every $L \in \mathbf{O}(\mathbb{R}^3)$ is $\operatorname{diag}(1, R_\theta)$ (rotation), $\operatorname{diag}(-1, 1, 1)$ (reflection) or $\operatorname{diag}(-1, R_\eta)$, $R_\eta \neq I$.

Definition 9.6.9 (Rotations of $\mathbb{R}^n$)

For every n, an element of $\mathbf{SO}(\mathbb{R}^n)$ (or of $\mathbf{SO}(n)$) is called a **rotation** of $\mathbb{R}^n$.

Example 9.6.10

The 4-dimensional rotation $\operatorname{diag}(R_\theta, R_\eta)$ has two "planes of rotation" with two angles.

All of $\mathbf{O}(\mathbb{R}^n)$ is described in Chapter 15.

Exercise 9.6.11

(a) $L \in \mathbf{O}(\mathbb{R}^n)$ iff $[L]^{\mathfrak{B}}_{\mathfrak{B}} \in \mathbf{O}(n)$ for some orthonormal basis $\mathfrak{B}$ of $\mathbb{R}^n$.
(b) The same holds with $\mathbf{SO}$ in place of $\mathbf{O}$.

Solution.

(a) $P = [I]^{\mathcal{E}}_{\mathfrak{B}} \in \mathbf{O}(n)$ and $[L]^{\mathfrak{B}}_{\mathfrak{B}} = P^{-1}[L]^{\mathcal{E}}_{\mathcal{E}}P$; $\mathbf{O}(n)$ is closed under products and inverses (Exercise 9.3.3).

(b) Additionally $\det[L]^{\mathfrak{B}}_{\mathfrak{B}} = \det L$. □